



Perturbative treatment of non-local operators in an ab initio context

Ryan Curry

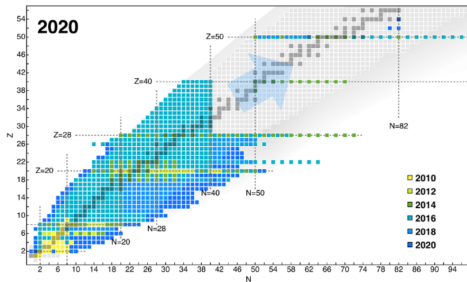
University of Guelph &
T2 Group at LANL

2024-08-22

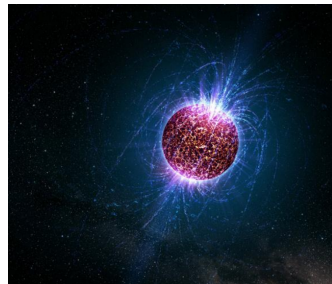
T2 Seminar
Los Alamos National Lab

Nuclear Systems

Nuclei

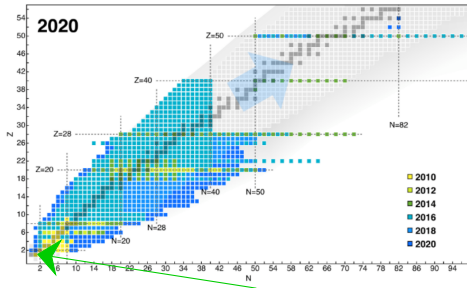


Neutron Stars

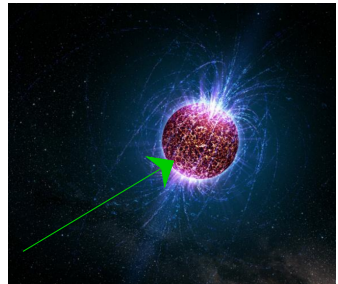


Nuclear Systems

Nuclei



Neutron Stars

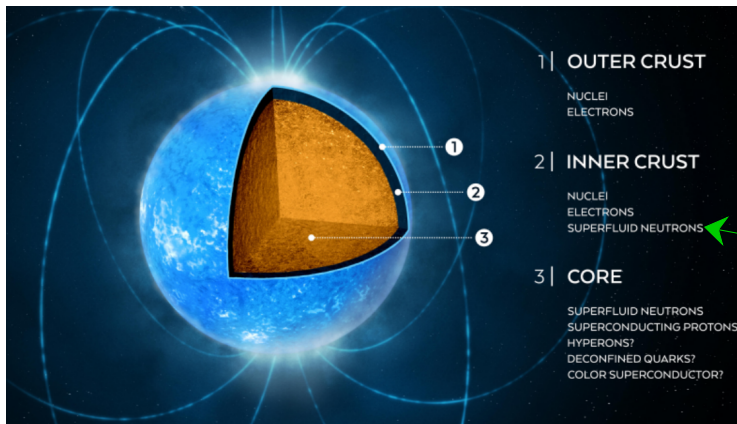


I will focus primarily on the deuteron, and the neutron matter equation of state.

Neutron Stars

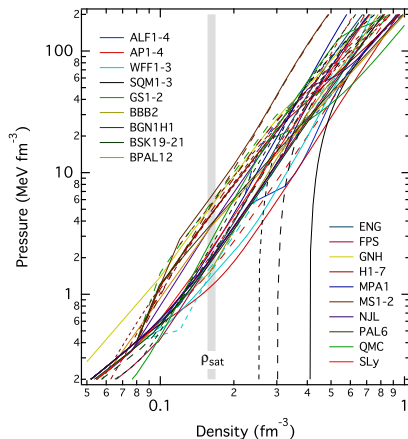
Radii $\sim 10 - 15$ km

Masses $\sim 1.4 M_{\odot} \leq 2.6 M_{\odot}$



Many-body calculations of neutron matter inform dense matter EOS

The Multi-Messenger Era



F.Özel and P. Freire

Annu. Rev. Astron. Astrophys. **54** (2016)

- Terrestrial experiments
 - Neutron skin thickness
 - Heavy ion collisions
- Astrophysical measurements
 - Gravitational Waves
 - NICER
 - Heavy pulsars
- Studies analyzing these events require input from nuclear theory
- We should do everything we can to reduce our theoretical uncertainties

Solving the Nuclear Many-Body Problem

**Accurate description of the
nuclear interaction**

+

**Powerful computational
approaches**

- Effective range expansion
- Argonne V18 & similar
 - Fit to experimental data
- Chiral EFT Interactions
 - Related to underlying theory (QCD)
- Hartree Fock (& extensions)
- BCS Theory
- Nuclear Shell Model
- Energy Density Functionals
- Many-Body Perturbation Theory
- Quantum Monte Carlo
- Coupled Cluster
- In-Medium Similarity Renormalization Group
- Self-Consistent Green's Functions

Chiral Effective Field Theory

	2N	3N	4N
LO $v = 0$			
NLO $v = 2$			
N ² LO $v = 3$			
N ³ LO $v = 4$			

- Expansion in powers of Q/Λ_b
- Separation of scales
quarks vs nucleons and
pions
- Power counting
- Many-body forces emerge
naturally
- Factor of 2 reduction in
error at each successive
chiral order!

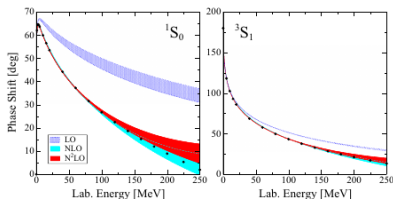
S. Weinberg, Phys. Lett. B, **251**, 288 (1990)

U. van Kolck, Phys. Rev. C, **49**, 2932 (1994)

E. Epelbaum, H.-W. Hammer, Ulf-G. Meissner,
Rev. Mod. Phys. **81**, 1773 (2009)

Local Chiral EFT

- Use only half of the operators in calculation
- Recover the rest through anti-symmetrization



- $q \rightarrow r$ local
- $k \rightarrow \nabla$ non-local

$$V^{(0)} = \alpha_1 + \alpha_2 \sigma_1 \cdot \sigma_2 + \alpha_3 \tau_1 \cdot \tau_2 + \alpha_4 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2$$

$$\begin{aligned} V_{\text{cont}}^{(2)} = & \gamma_1 q^2 + \gamma_2 q^2 \sigma_1 \cdot \sigma_2 + \gamma_3 q^2 \tau_1 \cdot \tau_2 \\ & + \gamma_4 q^2 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\ & + \gamma_5 k^2 + \gamma_6 k^2 \sigma_1 \cdot \sigma_2 + \gamma_7 k^2 \tau_1 \cdot \tau_2 \\ & + \gamma_8 k^2 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\ & + \gamma_9 (\sigma_1 + \sigma_2)(q \times k) \\ & + \gamma_{10} (\sigma_1 + \sigma_2)(q \times k) \tau_1 \cdot \tau_2 \\ & + \gamma_{11} (\sigma_1 \cdot q)(\sigma_2 \cdot q) \\ & + \gamma_{12} (\sigma_1 \cdot q)(\sigma_2 \cdot q) \tau_1 \cdot \tau_2 \\ & + \gamma_{13} (\sigma_1 \cdot k)(\sigma_2 \cdot k) \\ & + \gamma_{14} (\sigma_1 \cdot k)(\sigma_2 \cdot k) \tau_1 \cdot \tau_2 \end{aligned}$$

Trouble on the horizon ...

- 30 new contact operators at N³LO
- Cannot avoid non-local operators

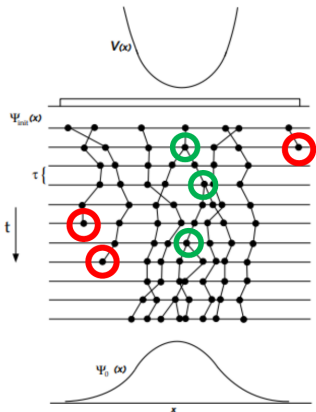
$$\begin{aligned}
 V_{\text{cont}}^{(4)} = & \alpha_1 q^4 + \alpha_2 q^4 \tau_1 \cdot \tau_2 + \alpha_3 q^4 \sigma_1 \cdot \sigma_2 + \alpha_4 q^4 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\
 & + \alpha_5 k^4 + \alpha_6 k^4 \tau_1 \cdot \tau_2 + \alpha_7 k^4 \sigma_1 \cdot \sigma_2 + \alpha_8 k^4 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\
 & + \alpha_9 q^2 k^2 + \alpha_{10} q^2 k^2 \tau_1 \cdot \tau_2 + \alpha_{11} q^2 k^2 \sigma_1 \cdot \sigma_2 + \alpha_{12} q^2 k^2 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\
 & + \alpha_{13} (q \times k)^2 + \alpha_{14} (q \times k)^2 \tau_1 \cdot \tau_2 + \alpha_{15} (q \times k)^2 \sigma_1 \cdot \sigma_2 + \alpha_{16} (q \times k)^2 \sigma_1 \cdot \sigma_2 \tau_1 \cdot \tau_2 \\
 & + \frac{i}{2} \alpha_{17} q^2 (\sigma_1 + \sigma_2) \cdot (q \times k) + \frac{i}{2} \alpha_{18} q^2 (\sigma_1 + \sigma_2) \cdot (q \times k) \tau_1 \cdot \tau_2 \\
 & + \frac{i}{2} \alpha_{19} k^2 (\sigma_1 + \sigma_2) \cdot (q \times k) + \frac{i}{2} \alpha_{20} k^2 (\sigma_1 + \sigma_2) \cdot (q \times k) \tau_1 \cdot \tau_2 \\
 & + \alpha_{21} q^2 \sigma_1 \cdot q \sigma_2 \cdot q + \alpha_{22} q^2 \sigma_1 \cdot q \sigma_2 \cdot q \tau_1 \cdot \tau_2 + \alpha_{23} k^2 \sigma_1 \cdot q \sigma_2 \cdot q + \alpha_{24} k^2 \sigma_1 \cdot q \sigma_2 \cdot q \tau_1 \cdot \tau_2 \\
 & + \alpha_{25} q^2 \sigma_1 \cdot k \sigma_2 \cdot k + \alpha_{26} q^2 \sigma_1 \cdot k \sigma_2 \cdot k \tau_1 \cdot \tau_2 + \alpha_{27} k^2 \sigma_1 \cdot k \sigma_2 \cdot k + \alpha_{28} k^2 \sigma_1 \cdot k \sigma_2 \cdot k \tau_1 \cdot \tau_2 \\
 & + \alpha_{29} \sigma_1 \cdot (q \times k) \sigma_2 \cdot (q \times k) + \alpha_{30} \sigma_1 \cdot (q \times k) \sigma_2 \cdot (q \times k) \tau_1 \cdot \tau_2.
 \end{aligned}$$

- Need develop a way to self consistently treat non-local operators perturbatively (more on that later)

Diffusion Monte Carlo

Schrodinger equation in imaginary time

$$-\frac{\partial}{\partial \tau} \Psi = \left(-\frac{\hbar}{2m} \nabla^2 + V \right) \Psi$$



$$-\frac{\partial}{\partial \tau} \Psi = \left(-\frac{\hbar}{2m} \nabla^2 + V \right) \Psi$$

Diffusion

$$-\frac{\partial}{\partial \tau} \Psi = \left(-\frac{\hbar}{2m} \nabla^2 + V \right) \Psi$$

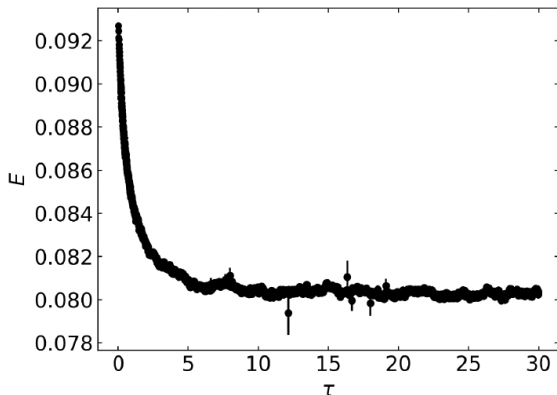
Growth / Decay

Diffusion Monte Carlo

$$\psi(\tau) = e^{-(H-E_T)\tau} \psi_T$$

$$\psi(\tau) = \sum_{i=0}^{\infty} a_i e^{-(E_i-E_T)\tau} \psi_i$$

$$\psi(\tau) = a_0 \psi_0 \quad \lim_{\tau \rightarrow \infty}$$



Spin / Isospin Complications

DMC is very good for central interactions

$$e^{-\sum_{i<j} v(\mathbf{r}_{ij})\tau}$$

How do we handle a propagator with spin/isospin degrees of freedom?

$$e^{-\sum_{i<j} [v_1(\mathbf{r}_{ij})\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + v_2(\mathbf{r}_{ij})\boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 + \dots]\tau}$$

The spin-spin term is equivalent to a Majorana exchange operator,

$$\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 = 2P_{12}^{\sigma} - 1$$

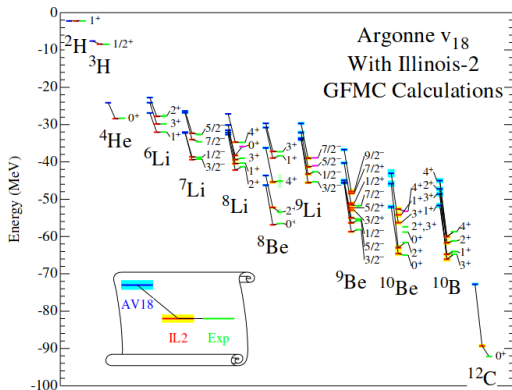
Green's Function Monte Carlo

- Explicitly sum over all spin/isospin states

$$|\Phi\rangle = \begin{pmatrix} a_{\uparrow\uparrow\uparrow} \\ a_{\uparrow\uparrow\downarrow} \\ a_{\uparrow\downarrow\uparrow} \\ a_{\uparrow\downarrow\downarrow} \\ a_{\downarrow\uparrow\uparrow} \\ a_{\downarrow\uparrow\downarrow} \\ a_{\downarrow\downarrow\uparrow} \\ a_{\downarrow\downarrow\downarrow} \end{pmatrix} \quad \sigma_1 \cdot \sigma_2 |\Phi\rangle = \begin{pmatrix} a_{\uparrow\uparrow\uparrow} \\ a_{\uparrow\uparrow\downarrow} \\ 2a_{\downarrow\uparrow\uparrow} - a_{\uparrow\downarrow\uparrow} \\ 2a_{\downarrow\uparrow\downarrow} - a_{\uparrow\downarrow\downarrow} \\ 2a_{\uparrow\downarrow\uparrow} - a_{\downarrow\uparrow\uparrow} \\ 2a_{\uparrow\downarrow\downarrow} - a_{\downarrow\uparrow\downarrow} \\ a_{\downarrow\downarrow\uparrow} \\ a_{\downarrow\downarrow\downarrow} \end{pmatrix}$$

- Each of the $2^A \frac{A!}{Z!(A-Z)!}$ spin-isospin states must be propagated
- Limiting GFMC to $A \leq 12$

Green's Function Monte Carlo



S. Pieper, R.B. Wiringa,
Annu. Rev. Nucl. Part. Sci. **51** (2001)

- GFMC is an exact calculation
- Limited to $A \leq 12$
- Still used to study electroweak observables,
G.B. King and S. Pastore,
Annu. Rev. Nucl. Part. Sci. **74**
(2024)
- and few-body nuclear reactions
J.E. Lynn, I. Tews, J. Carlson, *et. al*,
Phys. Rev. Lett., **116** (2016)

Auxiliary Field Diffusion Monte Carlo

- Instead of summing over all spin-isospin states, we can sample them
- Particles can be described by single-nucleon spinors

$$|S_i\rangle = a_i |p \uparrow\rangle + b_i |p \downarrow\rangle + c_i |n \uparrow\rangle + d_i |n \downarrow\rangle$$

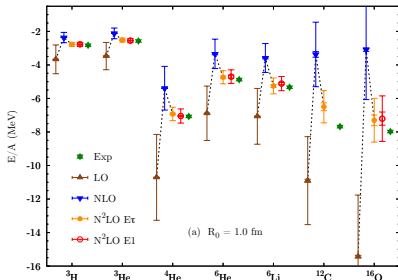
- Linearize quadratic spin/isospin ($\sigma_1 \cdot \sigma_2$, etc) dependence through Hubbard-Stratonovich Transformation

$$e^{-\frac{1}{2}\lambda O^2} = \frac{1}{\sqrt{2\pi}} \int dx_e e^{-\frac{x_e^2}{2} + \sqrt{-\lambda} x_e O}$$

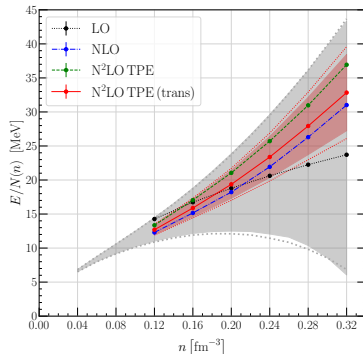
- We have traded two body $\sigma_1 \cdot \sigma_2$ type interactions for single particles interacting with external auxiliary fields x

Auxiliary Field Diffusion Monte Carlo

- Trade off in precision due to sampling auxiliary fields
- Extend calculations to $A \leq 18$ as well as dense matter



D. Lonardonì, S. Gandolfi, J.E. Lynn, *et. al*
Phys. Rev. C **97** (2018)



I. Tews, R. Somasundaram, *et. al*
arXiv:2407.08979 (2024)

What's the goal?

We have a non-perturbative many-body method, and a perturbative chiral EFT interaction.

Can we treat these interactions perturbatively?

R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, (2023)

At higher chiral orders, non-local operators cannot be avoided.

Can we come up with a self-consistent way to include them perturbatively in our many-body calculations?

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews,
In Preparation 2024

Perturbation Theory

First- and Second-order Corrections

$$E_0^{(1)} = \langle \psi_0 | V' | \psi_0 \rangle$$

$$E_0^{(2)} = - \sum_{k \neq 0}^{\infty} \frac{|\langle \psi_0 | V' | \psi_k \rangle|^2}{E_k - E_0}$$

The problem?

First- and Second-order Corrections

$$E_0^{(1)} = \langle \psi_0 | V' | \psi_0 \rangle$$

$$E_0^{(2)} = - \sum_{k \neq 0}^{\infty} \frac{|\langle \psi_0 | V' | \psi_k \rangle|^2}{E_k - E_0}$$

The problem?

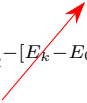
$$\lim_{\tau \rightarrow \infty} \psi(\tau) = \lim_{\tau \rightarrow \infty} e^{-(H-E_T)\tau} \psi_T \propto \psi_0 \longrightarrow E_0$$

A solution - In brief

$$I(\mathcal{T}) = \int_0^{\mathcal{T}} d\tau \langle \psi_0 | V' e^{-[H_0 - E_0]\tau} V' | \psi_0 \rangle$$

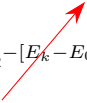
A solution - In brief

$$I(\mathcal{T}) = \int_0^{\mathcal{T}} d\tau \langle \psi_0 | V' e^{-[H_0 - E_0]\tau} V' | \psi_0 \rangle$$

$$I(\mathcal{T}) = (E_0^{(1)})^2 \mathcal{T} - \sum_{k \neq 0}^{\infty} \frac{|\langle \psi_k | V' | \psi_0 \rangle|^2}{E_k - E_0} [e^{-[E_k - E_0]\mathcal{T}} - 1]$$


A solution - In brief

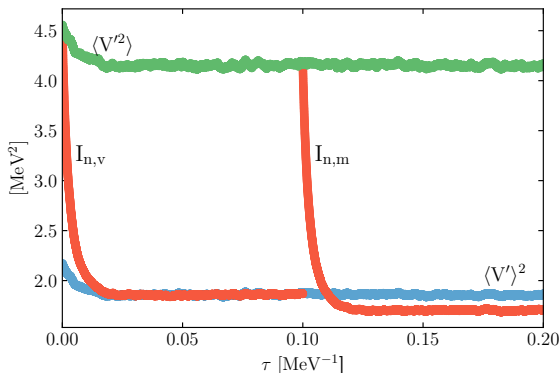
$$I(\mathcal{T}) = \int_0^{\mathcal{T}} d\tau \langle \psi_0 | V' e^{-[H_0 - E_0]\tau} V' | \psi_0 \rangle$$

$$I(\mathcal{T}) = (E_0^{(1)})^2 \mathcal{T} - \sum_{k \neq 0}^{\infty} \frac{|\langle \psi_k | V' | \psi_0 \rangle|^2}{E_k - E_0} [e^{-[E_k - E_0]\mathcal{T}} - 1]$$


$$I(\mathcal{T} \rightarrow \infty) = (E_0^{(1)})^2 \mathcal{T} - E_0^{(2)}$$

Practically speaking

$$I(\mathcal{T}) = \int_0^{\mathcal{T}} d\tau \langle \psi_0 | V' e^{-[H_0 - E_0]\tau} V' | \psi_0 \rangle \longrightarrow I \approx \frac{\sum_i^{\mathcal{N}} w_i V'(\mathbf{R}_i) V'(\mathbf{R}'_i)}{\sum_i^{\mathcal{N}} w_i}$$



R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, (2023)

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, *In Preparation* 2024

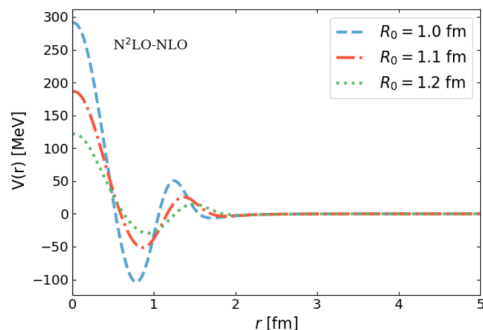
Testing Perturbativeness

$$V_{\text{chiral}} = V^{(0)} + V^{(2)} + V^{(2)} + \dots$$

LO NLO N²LO

66 Neutrons

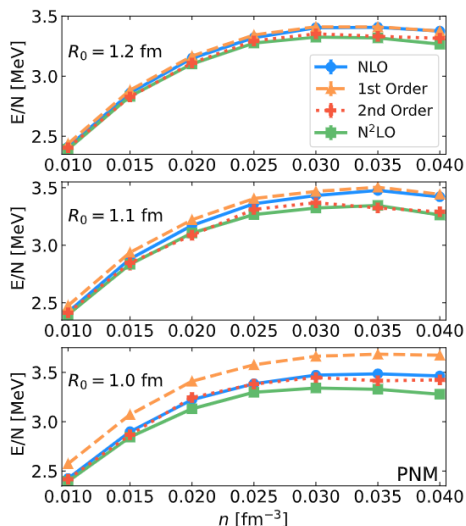
$$V' = \text{N}^2\text{LO} - \text{NLO}$$



Coordinate Space Cutoff R_0
 $\sim$ Potential Softness

Testing Perturbativeness

- Softest interaction ($R_0 = 1.2$ fm) has excellent agreement
- At ($R_0 = 1.0$ fm) there is no agreement between second-order and perturbative results
- Cast some doubt on perturbativeness of chiral EFT interactions.



Current Work

We have a non-perturbative many-body method, and a perturbative chiral EFT interaction.

Can we treat these interactions perturbatively?

R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, (2023)

At higher chiral orders, non-local operators cannot be avoided.

Can we come up with a self-consistent way to include them perturbatively in our many-body calculations?

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews,
In Preparation 2024

Non-Local Operators

- Non-local operators cannot be avoided at higher chiral orders.
- Non-local operators cannot be included in imaginary time propagator

$$V^{(0)} = \alpha_1 + \alpha_2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \alpha_3 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 + \alpha_4 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2$$

$$\begin{aligned} V_{\text{cont}}^{(2)} = & \gamma_1 q^2 + \gamma_2 q^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \gamma_3 q^2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_4 q^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_5 k^2 + \gamma_6 k^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \gamma_7 k^2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_8 k^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_9 (\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_2)(\mathbf{q} \times \mathbf{k}) \\ & + \gamma_{10} (\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_2)(\mathbf{q} \times \mathbf{k}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_{11} (\boldsymbol{\sigma}_1 \cdot \mathbf{q})(\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \\ & + \gamma_{12} (\boldsymbol{\sigma}_1 \cdot \mathbf{q})(\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_{13} (\boldsymbol{\sigma}_1 \cdot \mathbf{k})(\boldsymbol{\sigma}_2 \cdot \mathbf{k}) \\ & + \gamma_{14} (\boldsymbol{\sigma}_1 \cdot \mathbf{k})(\boldsymbol{\sigma}_2 \cdot \mathbf{k}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \end{aligned}$$

- $q \rightarrow r$ local
- $k \rightarrow \nabla$ non-local

Non-Local Operators

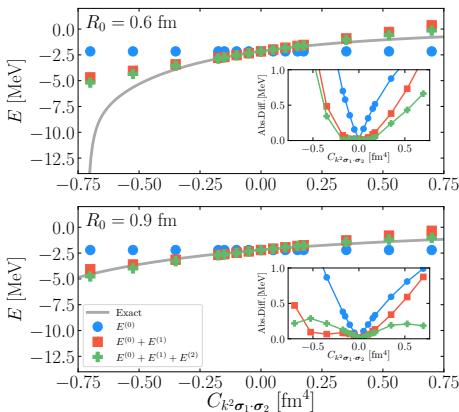
- Non-local operators cannot be avoided at higher chiral orders.
- Non-local operators cannot be included in imaginary time propagator
- Let's replace a local operator at NLO with a non-local equivalent and treat perturbatively
- $q \rightarrow r$ local
- $k \rightarrow \nabla$ non-local

$$V^{(0)} = \alpha_1 + \alpha_2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \alpha_3 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 + \alpha_4 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2$$

$$\begin{aligned} V_{\text{cont}}^{(2)} = & \gamma_1 q^2 + \gamma_2 q^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \gamma_3 q^2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_4 q^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_5 k^2 + \gamma_6 k^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \gamma_7 k^2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_8 k^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_9 (\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_2)(\mathbf{q} \times \mathbf{k}) \\ & + \gamma_{10} (\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_2)(\mathbf{q} \times \mathbf{k}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_{11} (\boldsymbol{\sigma}_1 \cdot \mathbf{q})(\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \\ & + \gamma_{12} (\boldsymbol{\sigma}_1 \cdot \mathbf{q})(\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \gamma_{13} (\boldsymbol{\sigma}_1 \cdot \mathbf{k})(\boldsymbol{\sigma}_2 \cdot \mathbf{k}) \\ & + \gamma_{14} (\boldsymbol{\sigma}_1 \cdot \mathbf{k})(\boldsymbol{\sigma}_2 \cdot \mathbf{k}) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \end{aligned}$$

Deuteron with non-local operator

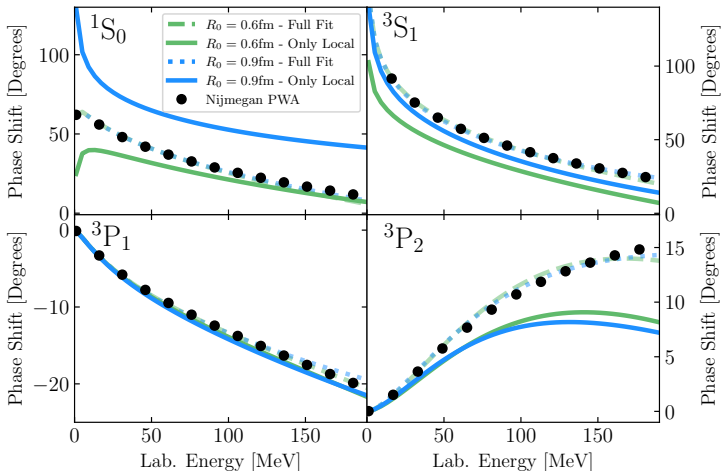
$$k^2 \longrightarrow \left[-\frac{1}{4}(\nabla^2 \delta_{R_0} \psi(\mathbf{r}) - \frac{1}{r} \frac{\partial \delta_{R_0}}{\partial r} (\mathbf{r} \cdot \nabla \psi(\mathbf{r})) - \delta_{R_0} \nabla^2 \psi(\mathbf{r}) \right]$$



- Non-local operator not included in fit to phase shifts
- $k^2 \sigma_1 \cdot \sigma_2$ included perturbatively for two cutoffs
- Compare against exact Lippmann-Schwinger solution

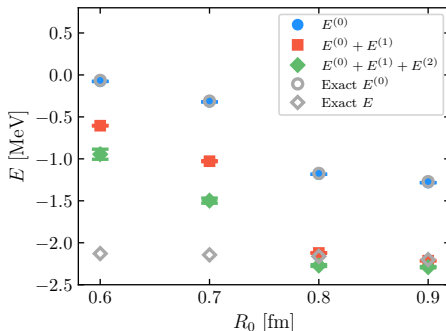
A realistic test case

- The non-local operator should be included in the fit to phase shifts



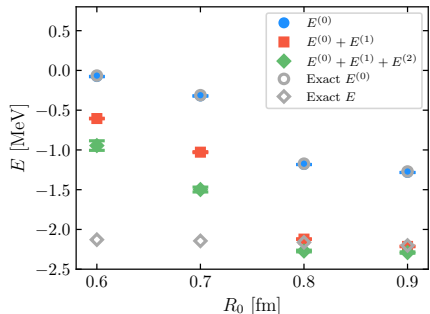
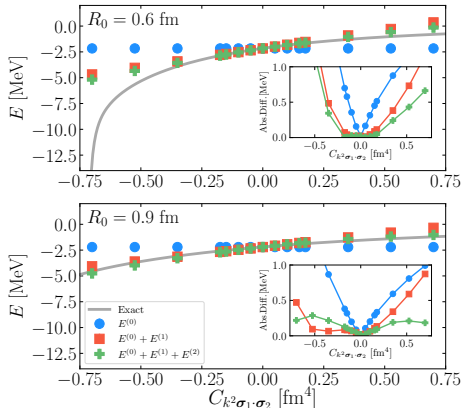
A realistic test case

$$k^2 \longrightarrow \left[-\frac{1}{4}(\nabla^2 \delta_{R_0} \psi(\mathbf{r}) - \frac{1}{r} \frac{\partial \delta_{R_0}}{\partial r} (\mathbf{r} \cdot \nabla \psi(\mathbf{r})) - \delta_{R_0} \nabla^2 \psi(\mathbf{r})) \right]$$



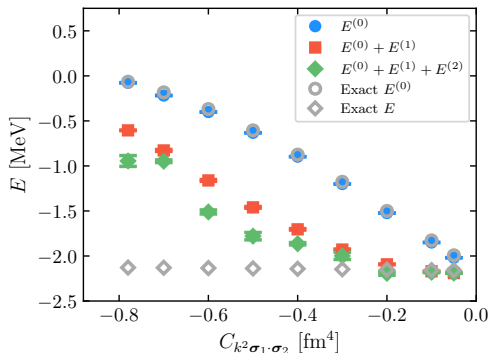
- Non-local operator included in fit to phase shifts
- Full range of coordinate space cutoffs
- Issues with perturbativeness at harder cutoffs

Wait a minute...



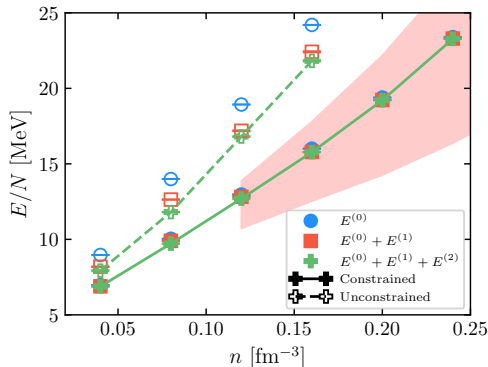
Can we take inspiration from the first case, and improve on the hard cutoff?

Constraining the non-local operator



- Non-local operator included in fit to phase shifts
- Create a series of NLO interactions with a variable non-local LEC
- Clear region where perturbation theory is sufficient!

Neutron Matter



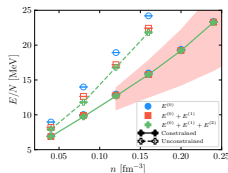
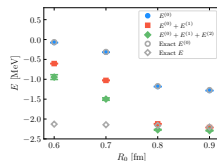
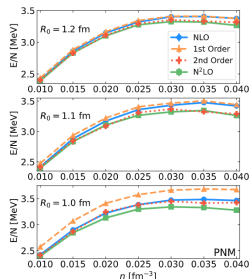
- Use constrained NLO interaction for calculation of neutron matter EOS
- Clear improvement from constraining the non-local operator
- Able to include non-local operators perturbatively even for hard cutoff interactions!

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, *In Preparation 2024*

I. Tews, R. Somasundaram, *et. al* arXiv:2407.08979

Summary

- Novel method for calculating second-order perturbation theory in an *ab initio* many-body context
- Treat non-local operators perturbatively for diverse nuclear systems
- Successfully constrained non-local operator to treat perturbatively for hard-cutoff interactions



Acknowledgements

Collaborators:

- Alex Gezerlis (Guelph)
- Ingo Tews (LANL)
- Stefano Gandolfi (LANL)
- Rahul Somasundaram (LANL, SU)
- Kevin Schmidt (ASU)
- Joel Lynn (Darmstadt)

Funding / Computational Resources:



Thank you for your attention!